

Comparative Efficacy of Carfilzomib-Based Versus Bortezomib-Based Regimens in Newly Diagnosed Multiple Myeloma: A Systematic Review and Meta-Analysis of Randomized Trials

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Purpose

To compare the efficacy of carfilzomib-based versus bortezomib-based regimens in patients with newly diagnosed multiple myeloma (NDMM), with a primary focus on progression-free survival (PFS) and secondary evaluation of overall response rate (ORR) and depth of response ($\geq$ complete response [CR]).

Background

Proteasome inhibitor-based triplet regimens are the cornerstone of frontline therapy for NDMM. Bortezomib-based regimens such as VRd remain widely adopted standards, while carfilzomib, a second-generation proteasome inhibitor with greater selectivity, has demonstrated the potential to induce deeper responses in regimens such as KRd. However, randomized trials comparing these approaches have yielded heterogeneous results, particularly regarding survival outcomes. A pooled analysis is necessary to clarify comparative efficacy and inform optimal frontline treatment selection.

Methods

A systematic literature search was conducted across PubMed-MEDLINE, Embase-OVID, CENTRAL, ClinicalTrials.gov, and Web of Science through March 1, 2026. Randomized phase III trials comparing carfilzomib-based with bortezomib-based regimens in NDMM were included. PFS was the primary outcome, while ORR and $\geq$ CR were secondary outcomes. Hazard ratios (HRs) for PFS and odds ratios (ORs) for response outcomes were extracted or derived from reported data. Pooled estimates were calculated using a random-effects inverse-variance model, with heterogeneity assessed using the I^2 statistic.

Results

Four randomized trials in NDMM were included. There was no statistically significant difference in PFS between carfilzomib-based and bortezomib-based regimens (HR 0.79; 95% CI, 0.48-1.27), with substantial heterogeneity ($I^2 = 69.6\%$). Carfilzomib-based regimens demonstrated a trend toward higher ORR (OR 1.26; 95% CI, 0.95-1.67; $I^2 = 0\%$). Notably, carfilzomib-based therapy was associated with significantly improved rates of deep response ($\geq$ CR) (OR 1.39; 95% CI, 1.04-1.85; $I^2 = 26.7\%$).

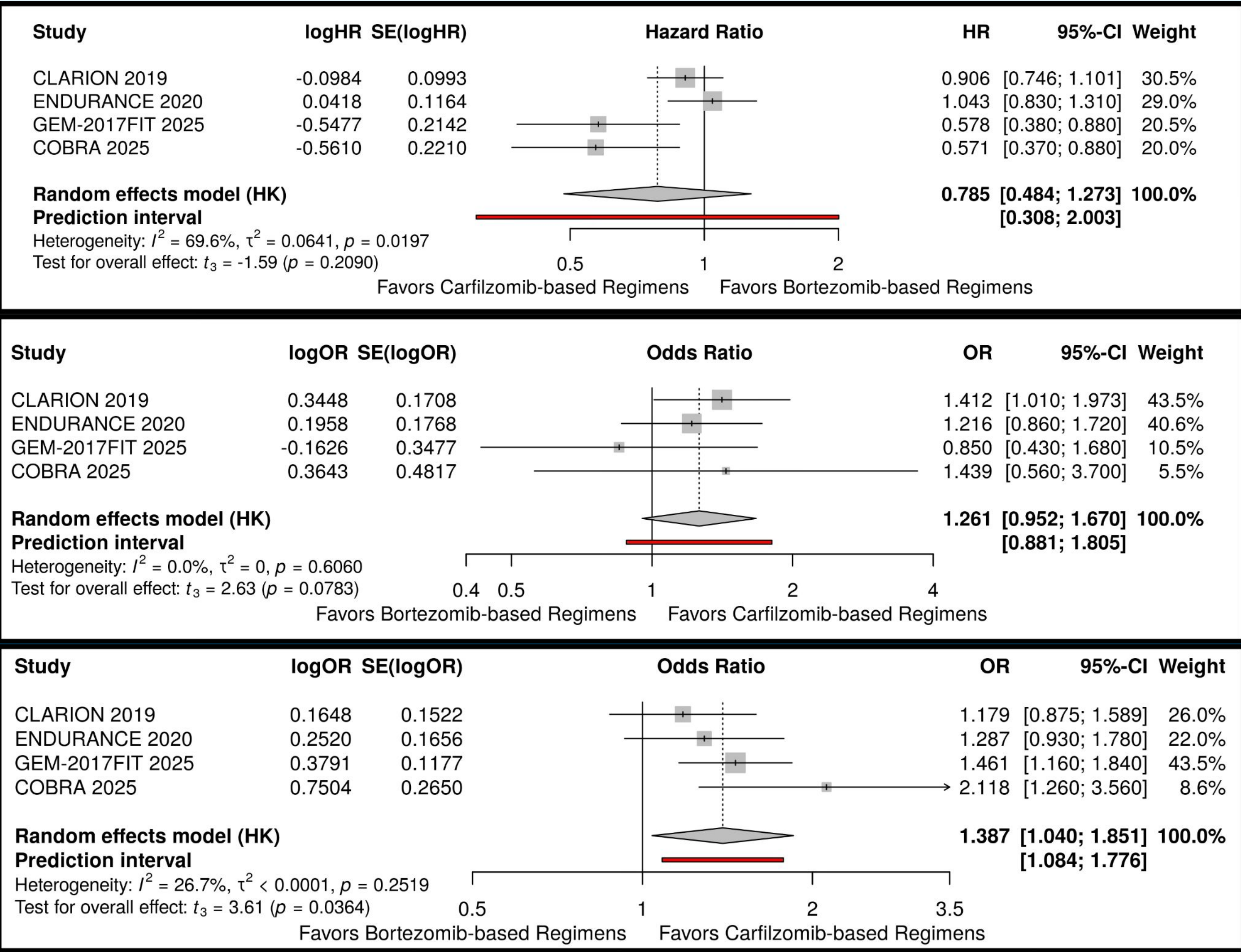


Figure 1. From top to bottom:
Pooled hazard ratios for progression-free survival (PFS)
Pooled odds ratios for overall response rate (ORR)
Pooled odds ratios for complete response or better ($\geq$ CR)

Implications

Carfilzomib-based regimens achieve significantly greater depth of response compared with bortezomib-based regimens in NDMM, although this advantage does not translate into a statistically significant PFS benefit. These findings suggest that deeper responses may not uniformly correlate with improved long-term outcomes and highlight the need for further studies.

